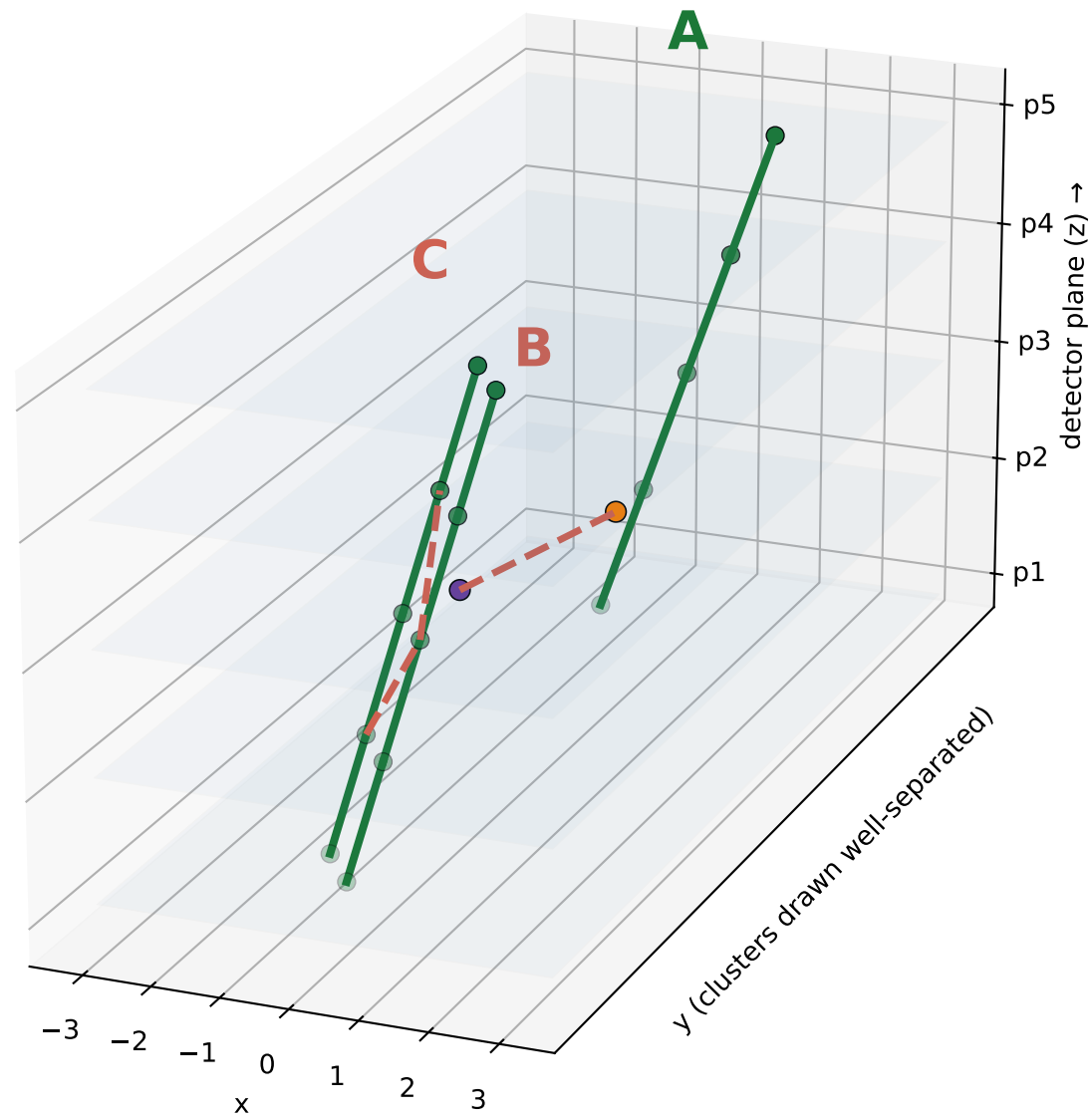
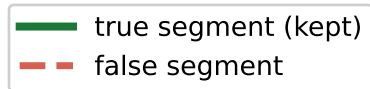


Schematic: the three segment types and what the 1-bit quantum filter does to each



A real track (4-segment chain): eigenvalues $\{2.4, 3.4, 4.6, 5.6\}$ are OFF the notch -> quantum KEEPS it (correct; keeps 3/4)
B isolated false segment (no compatible neighbour): eigenvalue = $\gamma + \delta = 4 = \text{the NOTCH}$ -> quantum ERASES it (correct)
C cross-track bridge (two near-parallel tracks accidentally aligned): eigenvalues OFF the notch -> quantum KEEPS it (FALSE POSITIVE)